

Lower apparent motor-unit discharge rates in simulated diabetic peripheral neuropathy reflect HD-sEMG-like selection rather than lower population rates

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Dependencies

Execution provenance

Item	Value
Executed at	2026-09-24T23:05:20-03:00
Python	3.12.7
marimo	0.23.16
NumPy	1.26.0
pandas	3.0.5
Matplotlib	3.11.1
SciPy	1.16.3
Git commit	fed72f6c4f4620fc6fddff0adfe11c52ec619783
Git state	clean
Notebook SHA-256	007d52d0f008dd5fc97bd1fc5316eb9b1b7d335f3b2eb75ab98b4946bf4cb3a8

Analysis configuration

Analysis helpers

All-active-MU simulation truth

For each simulated subject, the mean firing rate across every active motor unit is known exactly and defines the trial-specific simulation truth. This full-population value is the sole reference for the sampled HD-sEMG estimate.

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=== SIMULATION TRUTH (all motor units) ===
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FR simulation means (n=50 paired simulations/subjects): normal 13.00 ± 0.27
[95% BCa CI 12.93, 13.08]; DPN 14.04 ± 0.40 [95% BCa CI 13.93, 14.14]
Paired comparison (Simulation truth (all motor units)): DPN - normal = 1.03
pps [95% BCa CI 0.93, 1.13]; Wilcoxon signed-rank W=0.0, p=1.7764e-15
ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.217 ±
0.014 [95% BCa CI 0.213, 0.221]; DPN 0.150 ± 0.015 [95% BCa CI 0.146, 0.155]
Paired ISI-CoV comparison (Simulation truth (all motor units)): DPN - normal
= -0.067 [95% BCa CI -0.071, -0.063]; Wilcoxon signed-rank W=0.0, p=1.7764e-
15
Analyzed MUs per simulation/subject (mean ± SD [range]): normal 174.5 ± 2.9
[170, 182]; DPN 161.8 ± 3.1 [156, 171]
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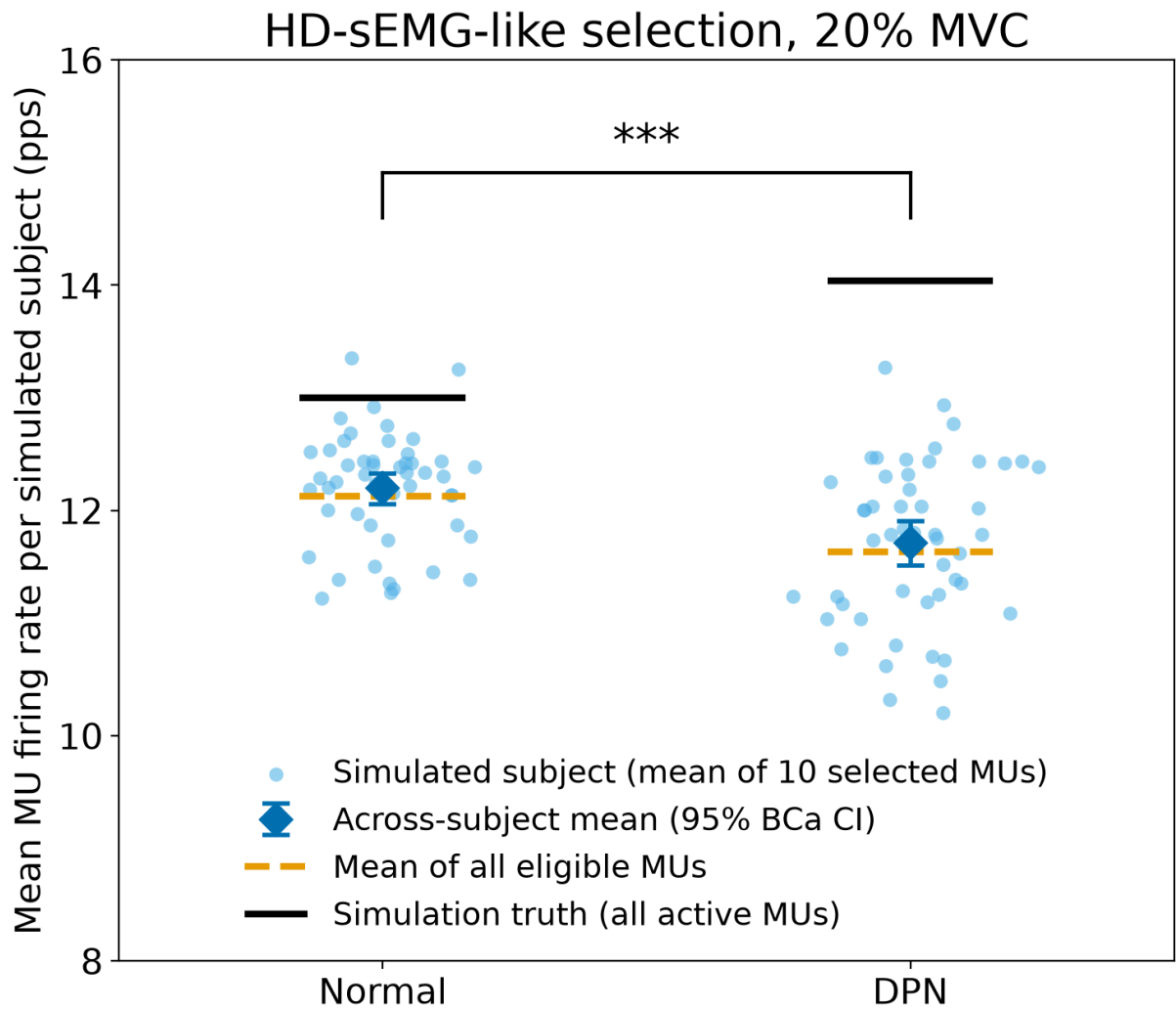
Firing rates at 20% MVC

Each simulation represents one subject. Group means, SDs, confidence intervals, and tests for firing rate and ISI-CoV therefore use one mean per simulation; raw motor-unit values are retained only for explicitly descriptive analyses. In the HD-sEMG mode, 10 different motor units are drawn at random from each subject-condition's pool of units meeting the firing-rate and ISI-CoV eligibility criteria (without replacement within each draw). One random-number stream per force level, seeded once, is consumed in trial order (Normal, then DPN), so the seed is not reset per subject or condition. Black lines show the condition means of the subject-specific simulation truths calculated from all active MUs; orange dashed lines show the condition means of all eligible MUs, the pools from which the 10-MU samples are drawn. Colours follow the Okabe-Ito palette, which is safe for red-green colour-vision deficiencies.

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=== HD-sEMG MODE (seeded random selection) ===
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Selection criteria: {'fmin': 5, 'fmax': 15, 'isicv': 0.3}
Motor units per simulation: 10
Selection seed: 20260102
FR simulation means (n=50 paired simulations/subjects): normal 12.20 ± 0.50
[95% BCa CI 12.06, 12.33]; DPN 11.71 ± 0.72 [95% BCa CI 11.51, 11.91]
Paired comparison (HD-sEMG): DPN - normal = -0.48 pps [95% BCa CI -0.73, -0.
25]; Wilcoxon signed-rank W=261.0, p=7.9659e-04
ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.153 ±
0.014 [95% BCa CI 0.149, 0.156]; DPN 0.126 ± 0.018 [95% BCa CI 0.121, 0.131]
Paired ISI-CoV comparison (HD-sEMG): DPN - normal = -0.027 [95% BCa CI -0.03
3, -0.020]; Wilcoxon signed-rank W=58.0, p=1.4433e-10
Analyzed MUs per simulation/subject (mean ± SD [range]): normal 10.0 ± 0.0
[10, 10]; DPN 10.0 ± 0.0 [10, 10]
=====
=== ELIGIBLE POOL (all HD-sEMG-eligible motor units) ===
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FR simulation means (n=50 paired simulations/subjects): normal 12.13 ± 0.30
[95% BCa CI 12.04, 12.21]; DPN 11.63 ± 0.34 [95% BCa CI 11.54, 11.73]
Paired comparison (Eligible pool): DPN - normal = -0.50 pps [95% BCa CI -0.6
1, -0.38]; Wilcoxon signed-rank W=60.0, p=1.8095e-10
ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.156 ±
0.008 [95% BCa CI 0.154, 0.158]; DPN 0.130 ± 0.009 [95% BCa CI 0.128, 0.133]
Paired ISI-CoV comparison (Eligible pool): DPN - normal = -0.026 [95% BCa CI
-0.029, -0.023]; Wilcoxon signed-rank W=0.0, p=1.7764e-15
Analyzed MUs per simulation/subject (mean ± SD [range]): normal 46.1 ± 5.6
[32, 57]; DPN 45.8 ± 5.7 [33, 61]

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Primary comparison figure saved to: diabetes/figures/mn_firing_rate_comparison_combined_v2.png

Statistical results saved to: diabetes/csv_results/mn_firing_rate_p_values_combined_v2.csv

Force-production statistics

Each simulation represents one subject and contributes one steady-state mean force and force coefficient of variation (CoV) per group. Normal-DPN estimates use paired simulation IDs, with 95% BCa bootstrap intervals and two-sided Wilcoxon signed-rank p-values.

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=== FORCE PRODUCTION STATISTICS (20% MVC, batch: variability) ===
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(Steady-state: t > 4000 ms)
(Bootstrap resamples: 100,000)

--- normal (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 59.9939 ± 0.0484 [95% BCa CI: 59.9815, 60.0081]
  Force as %MVC: 20.00%
  CoV of force: 0.0212 ± 0.0030 [95% BCa CI: 0.0205, 0.0222]

--- DPN (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 60.0113 ± 0.0572 [95% BCa CI: 59.9945, 60.0259]
  Force as %MVC: 20.00%
  CoV of force: 0.0147 ± 0.0028 [95% BCa CI: 0.0140, 0.0155]

--- Paired comparison (DPN - normal) ---
  Mean-force difference: 0.0173 [95% BCa CI: -0.0034, 0.0360]; paired Wilco
on W=445.0, p=6.3466e-02
  Force-CoV difference: -0.0065 [95% BCa CI: -0.0075, -0.0055]; paired Wilco
xon W=3.0, p=8.8818e-15

--- Between-simulation/subject variability ---
  normal: CoV of mean force across simulations/subjects = 0.08%
  normal: CoV of force CoV across simulations/subjects = 14.22%
  DPN: CoV of mean force across simulations/subjects = 0.10%
  DPN: CoV of force CoV across simulations/subjects = 18.73%

=====
=== FORCE PRODUCTION STATISTICS (10% MVC, batch: mvc10) ===
=====
(Steady-state: t > 4000 ms)
(Bootstrap resamples: 100,000)

--- normal (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 30.0396 ± 0.0398 [95% BCa CI: 30.0289, 30.0508]
  Force as %MVC: 10.01%
  CoV of force: 0.0274 ± 0.0048 [95% BCa CI: 0.0261, 0.0287]

--- DPN (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 30.0197 ± 0.0445 [95% BCa CI: 30.0070, 30.0313]
  Force as %MVC: 10.01%
  CoV of force: 0.0182 ± 0.0036 [95% BCa CI: 0.0173, 0.0192]

--- Paired comparison (DPN - normal) ---
  Mean-force difference: -0.0200 [95% BCa CI: -0.0350, -0.0052]; paired Wilc
oxon W=383.0, p=1.3305e-02
  Force-CoV difference: -0.0091 [95% BCa CI: -0.0110, -0.0075]; paired Wilco
xon W=10.0, p=7.6383e-14

--- Between-simulation/subject variability ---
  normal: CoV of mean force across simulations/subjects = 0.13%
  normal: CoV of force CoV across simulations/subjects = 17.60%
  DPN: CoV of mean force across simulations/subjects = 0.15%
  DPN: CoV of force CoV across simulations/subjects = 19.63%

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=====
=== FORCE PRODUCTION STATISTICS (50% MVC, batch: mvc50) ===
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(Steady-state: t > 4000 ms)
(Bootstrap resamples: 100,000)

--- normal (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 150.0022 ± 0.0788 [95% BCa CI: 149.9796, 150.0230]
  Force as %MVC: 50.00%
  CoV of force: 0.0159 ± 0.0019 [95% BCa CI: 0.0155, 0.0165]

--- DPN (n=50 paired simulations/subjects; 50 available) ---
  Mean force: 150.0128 ± 0.0706 [95% BCa CI: 149.9948, 150.0336]
  Force as %MVC: 50.00%
  CoV of force: 0.0106 ± 0.0017 [95% BCa CI: 0.0101, 0.0110]

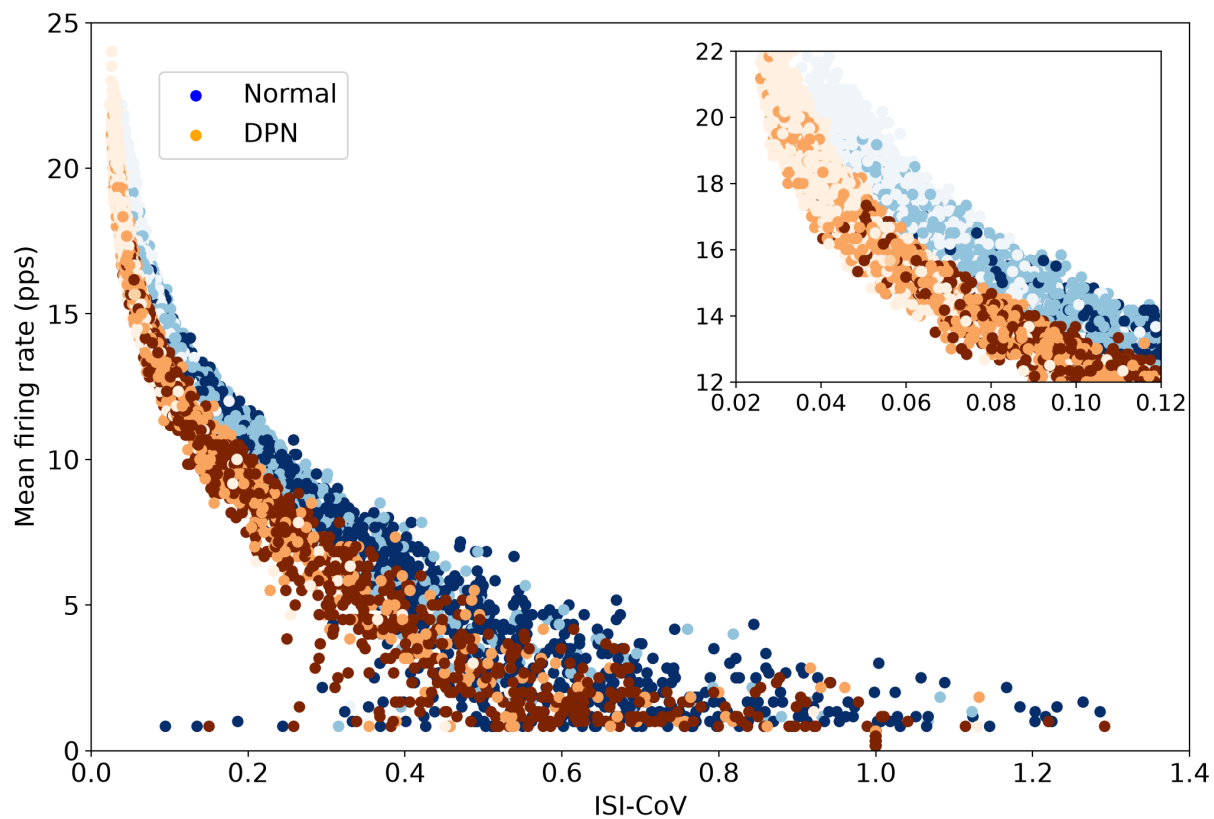
--- Paired comparison (DPN - normal) ---
  Mean-force difference: 0.0106 [95% BCa CI: -0.0181, 0.0419]; paired Wilcoxon W=598.0, p=7.0895e-01
  Force-CoV difference: -0.0054 [95% BCa CI: -0.0060, -0.0048]; paired Wilcoxon W=0.0, p=1.7764e-15

--- Between-simulation/subject variability ---
  normal: CoV of mean force across simulations/subjects = 0.05%
  normal: CoV of force CoV across simulations/subjects = 11.67%
  DPN: CoV of mean force across simulations/subjects = 0.05%
  DPN: CoV of force CoV across simulations/subjects = 16.50%

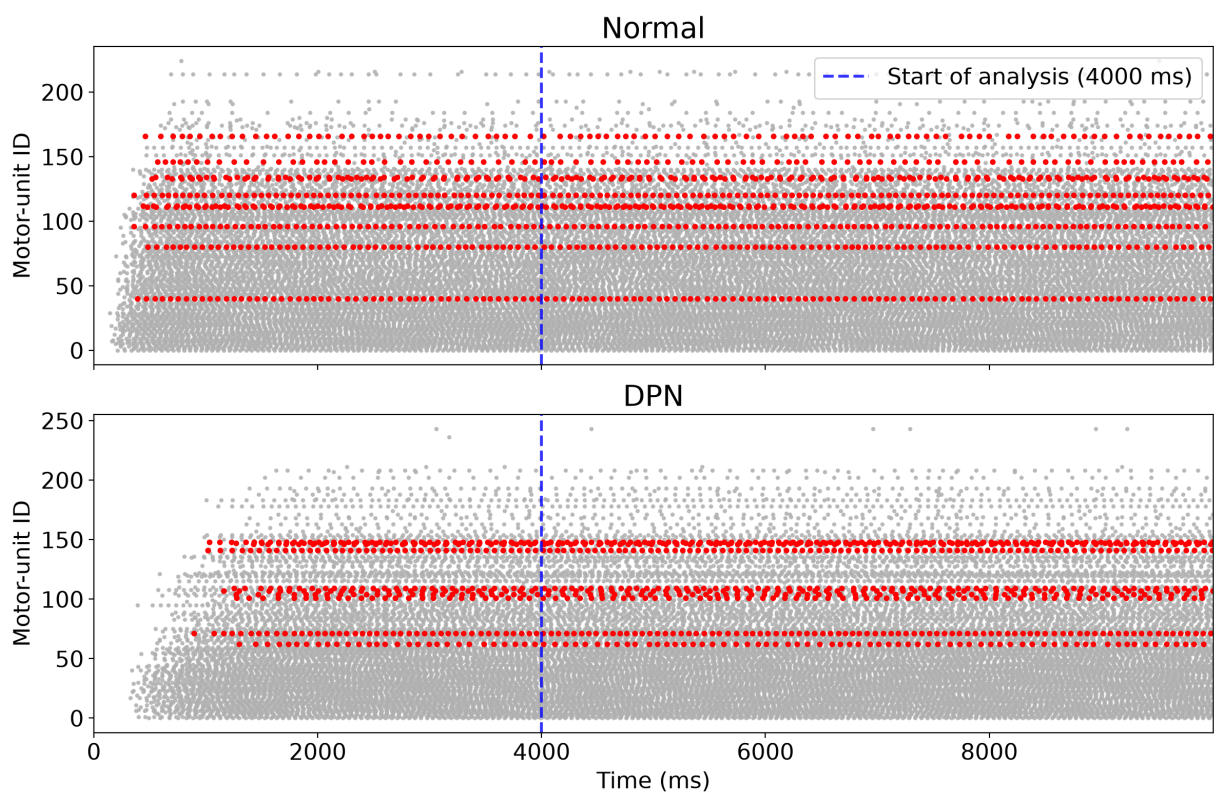
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Firing-rate and ISI-variability relationship

This scatterplot is an explicitly descriptive motor-unit-level view; it is not used for group-level inference.



Motor-unit selection visualizations

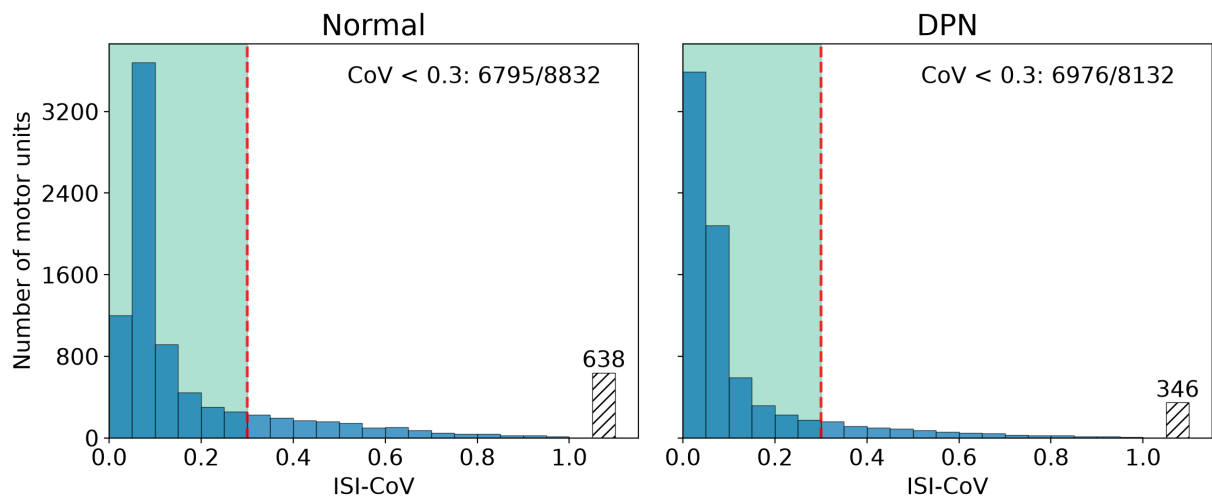


Motor-unit selection figure saved to: diabetes/figures/what_mn_selected_combined_trial_30_v2.png

Condition normal: min index=6, max index=210; mean ID span= 107.6 ± 22.2
Condition DPN: min index=4, max index=247; mean ID span= 127.4 ± 34.8
Paired subjects with larger DPN ID span: 33/50
Selected-MU ID summary saved to: diabetes/csv_results/hdsemg_selected_motor_units_v2.csv

ISI-variability distributions

These histograms describe individual motor units pooled across simulations; they are not used for subject-level estimates or tests.



ISI CoV histograms for all motor units saved to: diabetes/figures/isi_cov_histograms_all_units_variability_v2.png

Robustness across contraction intensities

The 20% MVC condition was designated as the primary reference analysis. The same simulation-level analysis was repeated at 10% and 50% MVC using 50 paired simulated subjects per force level to assess whether the direction and magnitude of the findings were robust across contraction intensities.

For firing rate and ISI-CoV, the analysis estimates the paired DPN-minus-Normal effect at each MVC, quantifies the difference between the randomized HD-sEMG estimate and the all-active-MU simulation truth, and tests force-level dependence using a within-subject Friedman test followed by all three paired MVC contrasts with Holm adjustment. Trial identifiers must match within each condition pair and across contraction intensities.

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=== FIRING RATES AT 10% MVC (batch: mvc10) ===
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--- Randomized HD-sEMG selection ---

Selection criteria: {'fmin': 5, 'fmax': 15, 'isicv': 0.3}

Selection seed: 20260104

FR simulation means (n=50 paired simulations/subjects): normal 12.03 ± 0.66 [95% BCa CI 11.85, 12.21]; DPN 11.76 ± 0.80 [95% BCa CI 11.54, 11.98]

Paired comparison (10% MVC HD-sEMG): DPN - normal = -0.27 pps [95% BCa CI -0.59, 0.01]; Wilcoxon signed-rank $W=479.5$, $p=1.8581e-01$

ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.167 ± 0.020 [95% BCa CI 0.162, 0.172]; DPN 0.137 ± 0.023 [95% BCa CI 0.131, 0.143]

Paired ISI-CoV comparison (10% MVC HD-sEMG): DPN - normal = -0.030 [95% BCa CI -0.038, -0.021]; Wilcoxon signed-rank $W=94.0$, $p=5.3101e-09$

Analyzed MUs per simulation/subject (mean $\pm$ SD [range]): normal 10.0 ± 0.0 [10, 10]; DPN 10.0 ± 0.0 [10, 10]

--- Simulation truth (all active MUs) ---

FR simulation means (n=50 paired simulations/subjects): normal 10.13 ± 0.39 [95% BCa CI 10.02, 10.24]; DPN 11.52 ± 0.44 [95% BCa CI 11.39, 11.64]

Paired comparison (10% MVC all motor units): DPN - normal = 1.39 pps [95% BCa CI 1.22, 1.54]; Wilcoxon signed-rank $W=0.0$, $p=1.7764e-15$

ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.321 ± 0.021 [95% BCa CI 0.315, 0.327]; DPN 0.228 ± 0.023 [95% BCa CI 0.221, 0.234]

Paired ISI-CoV comparison (10% MVC all motor units): DPN - normal = -0.093 [95% BCa CI -0.102, -0.084]; Wilcoxon signed-rank $W=0.0$, $p=1.7764e-15$

Analyzed MUs per simulation/subject (mean $\pm$ SD [range]): normal 138.6 ± 3.7 [129, 147]; DPN 123.3 ± 3.0 [118, 131]

--- Eligible pool (all HD-sEMG-eligible MUs) ---

FR simulation means (n=50 paired simulations/subjects): normal 12.10 ± 0.29 [95% BCa CI 12.02, 12.18]; DPN 11.71 ± 0.32 [95% BCa CI 11.62, 11.80]

Paired comparison (10% MVC eligible pool): DPN - normal = -0.39 pps [95% BCa CI -0.51, -0.28]; Wilcoxon signed-rank $W=148.0$, $p=3.4864e-07$

ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.165 ± 0.009 [95% BCa CI 0.163, 0.168]; DPN 0.138 ± 0.009 [95% BCa CI 0.136, 0.141]

Paired ISI-CoV comparison (10% MVC eligible pool): DPN - normal = -0.027 [95% BCa CI -0.030, -0.023]; Wilcoxon signed-rank $W=1.0$, $p=3.5527e-15$

Analyzed MUs per simulation/subject (mean $\pm$ SD [range]): normal 49.5 ± 5.0 [39, 64]; DPN 48.9 ± 6.3 [34, 65]

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=== FIRING RATES AT 50% MVC (batch: mvc50) ===
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--- Randomized HD-sEMG selection ---

Selection criteria: {'fmin': 5, 'fmax': 15, 'isicv': 0.3}

Selection seed: 20260105

FR simulation means (n=50 paired simulations/subjects): normal 11.90 ± 0.72 [95% BCa CI 11.71, 12.10]; DPN 11.49 ± 0.90 [95% BCa CI 11.23, 11.73]

Paired comparison (50% MVC HD-sEMG): DPN - normal = -0.41 pps [95% BCa CI -0.74, -0.11]; Wilcoxon signed-rank $W=424.0$, $p=3.9299e-02$

ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.141 ± 0.019 [95% BCa CI 0.136, 0.147]; DPN 0.125 ± 0.021 [95% BCa CI 0.120, 0.131]

Paired ISI-CoV comparison (50% MVC HD-sEMG): DPN - normal = -0.016 [95% BCa CI -0.024, -0.008]; Wilcoxon signed-rank W=272.0, p=2.7386e-04
Analyzed MUs per simulation/subject (mean $\pm$ SD [range]): normal 10.0 $\pm$ 0.0 [10, 10]; DPN 10.0 $\pm$ 0.0 [10, 10]

--- Simulation truth (all active MUs) ---

FR simulation means (n=50 paired simulations/subjects): normal 16.88 $\pm$ 0.23 [95% BCa CI 16.82, 16.95]; DPN 17.42 $\pm$ 0.38 [95% BCa CI 17.31, 17.52]
Paired comparison (50% MVC all motor units): DPN - normal = 0.53 pps [95% BCa CI 0.43, 0.64]; Wilcoxon signed-rank W=6.0, p=2.4869e-14
ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.115 $\pm$ 0.008 [95% BCa CI 0.113, 0.117]; DPN 0.085 $\pm$ 0.010 [95% BCa CI 0.082, 0.087]
Paired ISI-CoV comparison (50% MVC all motor units): DPN - normal = -0.030 [95% BCa CI -0.034, -0.027]; Wilcoxon signed-rank W=0.0, p=1.7764e-15
Analyzed MUs per simulation/subject (mean $\pm$ SD [range]): normal 228.4 $\pm$ 2.1 [223, 233]; DPN 221.2 $\pm$ 2.4 [215, 225]

--- Eligible pool (all HD-sEMG-eligible MUs) ---

FR simulation means (n=50 paired simulations/subjects): normal 11.77 $\pm$ 0.37 [95% BCa CI 11.67, 11.88]; DPN 11.56 $\pm$ 0.41 [95% BCa CI 11.44, 11.66]
Paired comparison (50% MVC eligible pool): DPN - normal = -0.21 pps [95% BCa CI -0.38, -0.06]; Wilcoxon signed-rank W=382.0, p=1.2932e-02
ISI-CoV simulation means (n=50 paired simulations/subjects): normal 0.145 $\pm$ 0.010 [95% BCa CI 0.143, 0.148]; DPN 0.124 $\pm$ 0.010 [95% BCa CI 0.122, 0.127]
Paired ISI-CoV comparison (50% MVC eligible pool): DPN - normal = -0.021 [95% BCa CI -0.025, -0.017]; Wilcoxon signed-rank W=35.0, p=7.6561e-12
Analyzed MUs per simulation/subject (mean $\pm$ SD [range]): normal 37.6 $\pm$ 4.5 [30, 49]; DPN 38.7 $\pm$ 5.0 [30, 50]

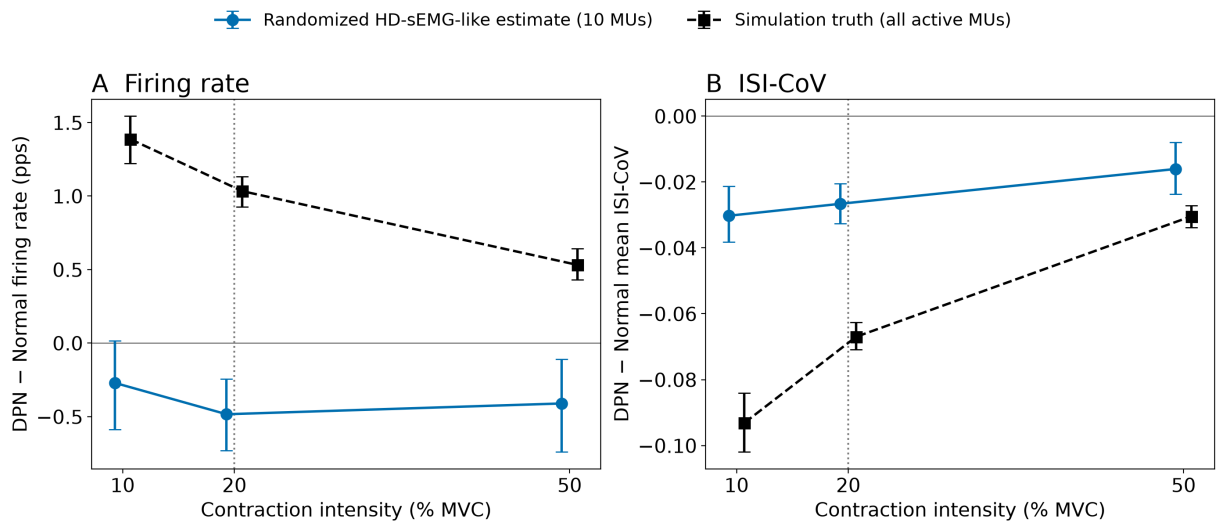
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=== FORCE-LEVEL ROBUSTNESS OF DPN - NORMAL EFFECTS ===
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firing_rate (pps)
  10% MVC, randomized_hdsemg: -0.2707 [-0.5880, +0.0143], p=0.1858, Holm p=
0.1858
  20% MVC, randomized_hdsemg: -0.4840 [-0.7320, -0.2450], p=0.0007966, Holm
p=0.00239
  50% MVC, randomized_hdsemg: -0.4107 [-0.7407, -0.1100], p=0.0393, Holm p=
0.0786
  10% MVC, simulation_truth_all_active_motor_units: +1.3862 [+1.2203, +1.543
0], p=1.776e-15, Holm p=5.329e-15
  20% MVC, simulation_truth_all_active_motor_units: +1.0331 [+0.9252, +1.133
4], p=1.776e-15, Holm p=5.329e-15
  50% MVC, simulation_truth_all_active_motor_units: +0.5314 [+0.4294, +0.642
8], p=2.487e-14, Holm p=2.487e-14
  MVC dependence, randomized_hdsemg: Friedman Q=4.120, p=0.1275
    10_minus_20: +0.2133 [-0.2013, +0.5953], Holm p=0.5775
    50_minus_20: +0.0733 [-0.3460, +0.4923], Holm p=1
    50_minus_10: -0.1400 [-0.5517, +0.2603], Holm p=1
  MVC dependence, simulation_truth_all_active_motor_units: Friedman Q=36.28
0, p=1.324e-08
    10_minus_20: +0.3530 [+0.1781, +0.5113], Holm p=0.0001702
    50_minus_20: -0.5018 [-0.6429, -0.3455], Holm p=1.681e-07
    50_minus_10: -0.8548 [-1.0601, -0.6361], Holm p=8.439e-09

isi_cv (unitless)
  10% MVC, randomized_hdsemg: -0.0302 [-0.0383, -0.0213], p=5.31e-09, Holm p
=1.062e-08
  20% MVC, randomized_hdsemg: -0.0266 [-0.0327, -0.0204], p=1.443e-10, Holm
p=4.33e-10
  50% MVC, randomized_hdsemg: -0.0160 [-0.0236, -0.0079], p=0.0002739, Holm
p=0.0002739
  10% MVC, simulation_truth_all_active_motor_units: -0.0931 [-0.1020, -0.084
0], p=1.776e-15, Holm p=5.329e-15
  20% MVC, simulation_truth_all_active_motor_units: -0.0670 [-0.0709, -0.062
6], p=1.776e-15, Holm p=5.329e-15
  50% MVC, simulation_truth_all_active_motor_units: -0.0304 [-0.0338, -0.027
1], p=1.776e-15, Holm p=5.329e-15
  MVC dependence, randomized_hdsemg: Friedman Q=5.880, p=0.05287
    10_minus_20: -0.0036 [-0.0141, +0.0078], Holm p=0.323
    50_minus_20: +0.0106 [+0.0006, +0.0202], Holm p=0.09662
    50_minus_10: +0.0142 [+0.0033, +0.0255], Holm p=0.04594
  MVC dependence, simulation_truth_all_active_motor_units: Friedman Q=61.72
0, p=3.96e-14
    10_minus_20: -0.0261 [-0.0363, -0.0158], Holm p=1.976e-05
    50_minus_20: +0.0366 [+0.0313, +0.0416], Holm p=2.665e-14
    50_minus_10: +0.0627 [+0.0522, +0.0723], Holm p=4.867e-13

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Force-level robustness figure saved to: diabetes/figures/force_level_robustness_v2.png

Statistical summaries saved to diabetes/csv_results/force_level*_v2.csv

Selection-threshold sensitivity analysis

The ISI-CoV ceiling is the least standardized eligibility criterion (0.25-0.50 in the cited studies), and a regularity criterion acts differently on populations whose discharge variability differs. At 10%, 20%, and 50% MVC, the ceiling was therefore varied from 0.15 to 0.50 while holding the firing-rate window and sample size constant. Whereas the selection-seed analysis varies only which eligible MUs are drawn, this analysis varies the eligibility criterion itself; neither contributes to the primary fixed-seed estimates.

The HD-sEMG analysis first applies a firing-rate window ($f_{\min} < FR < f_{\max}$) and an ISI-CoV ceiling, then randomly samples 10 eligible motor units without replacement. Per-MU firing rates and ISI-CoV values are computed once per simulated subject and reused across all analyses below.

Selection stability is evaluated over 1,000 prespecified seeds at 20%, 10%, and 50% MVC. The threshold analysis uses that same ordered seed set at every ISI-CoV threshold (common random numbers), so differences along the curve are less affected by unrelated draw noise. Its shaded band is the central 95% across-selection-seed range, not a confidence interval. The mean of all eligible MUs (open orange triangles) is the value each 10-MU sample estimates, and the known mean over all active motor units is shown separately as the simulation truth.

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=== RANDOMIZED HD-sEMG SELECTION-STABILITY ANALYSIS ===
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20% MVC (50 pairs): fixed seed 20260102, difference -0.484 pps, p=0.0007966
  Across-selection-seed range: -0.743 to -0.267 pps; median -0.495
  Eligible pool (minimum/median/maximum): 32/46.0/61
  Simulation truth: +1.033 pps, p=1.776e-15
10% MVC (50 pairs): fixed seed 20260104, difference -0.271 pps, p=0.1858
  Across-selection-seed range: -0.630 to -0.163 pps; median -0.398
  Eligible pool (minimum/median/maximum): 34/49.0/65
  Simulation truth: +1.386 pps, p=1.776e-15
50% MVC (50 pairs): fixed seed 20260105, difference -0.411 pps, p=0.0393
  Across-selection-seed range: -0.472 to +0.064 pps; median -0.205
  Eligible pool (minimum/median/maximum): 30/37.0/50
  Simulation truth: +0.531 pps, p=2.487e-14
=====
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=== SELECTION-THRESHOLD SENSITIVITY (10%, 20%, 50% MVC) ===
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Fixed: fmin=5 pps, fmax=15 pps, mn_number=10, 50 paired subjects per force level
Common ordered seed set: 20261000-20261999 at every threshold and force level

10% MVC - simulation truth (all active MUs): +1.39 pps (Normal 10.13, DPN 11.52)
  ISI-CoV    median difference          95% seed range  eligible pool  min
pool
    0.15      -0.631      -0.731 to -0.533      -0.635      15
    0.20      -0.542      -0.691 to -0.392      -0.543      23
    0.25      -0.480      -0.668 to -0.288      -0.482      29
    0.30      -0.398      -0.630 to -0.163      -0.393      34
<- main analysis
    0.32      -0.379      -0.624 to -0.106      -0.372      36
    0.34      -0.322      -0.577 to -0.053      -0.323      39
    0.36      -0.301      -0.567 to -0.001      -0.300      41
    0.38      -0.219      -0.491 to +0.053      -0.220      42
    0.40      -0.166      -0.464 to +0.125      -0.160      42
    0.45       +0.013      -0.294 to +0.317       +0.004      44
    0.50       +0.107      -0.199 to +0.416       +0.103      44

20% MVC - simulation truth (all active MUs): +1.03 pps (Normal 13.00, DPN 14.04)
  ISI-CoV    median difference          95% seed range  eligible pool  min
pool
    0.15      -0.607      -0.724 to -0.486      -0.607      19
    0.20      -0.601      -0.750 to -0.449      -0.602      23
    0.25      -0.586      -0.787 to -0.377      -0.583      27
    0.30      -0.495      -0.743 to -0.267      -0.497      32
<- main analysis
    0.32      -0.440      -0.681 to -0.187      -0.445      34
    0.34      -0.364      -0.643 to -0.107      -0.370      34

```

0.36	-0.311	-0.604 to	-0.040	-0.312	34
0.38	-0.203	-0.486 to	+0.076	-0.203	34
0.40	-0.149	-0.442 to	+0.135	-0.140	35
0.45	+0.048	-0.258 to	+0.350	+0.047	35
0.50	+0.125	-0.175 to	+0.429	+0.124	35

50% MVC - simulation truth (all active MUs): +0.53 pps (Normal 16.88, DPN 1 7.42)

ISI-CoV pool	median difference	95% seed range	eligible pool	min
0.15	-0.448	-0.564 to -0.326	-0.449	17
0.20	-0.379	-0.562 to -0.211	-0.383	22
0.25	-0.339	-0.574 to -0.103	-0.337	27
0.30	-0.205	-0.472 to +0.064	-0.215	30

<- main analysis

0.32	-0.166	-0.436 to	+0.097	-0.174	30
0.34	-0.103	-0.391 to	+0.187	-0.108	30
0.36	-0.071	-0.364 to	+0.235	-0.067	30
0.38	-0.024	-0.311 to	+0.272	-0.024	30
0.40	+0.029	-0.264 to	+0.336	+0.025	30
0.45	+0.095	-0.198 to	+0.391	+0.097	30
0.50	+0.119	-0.182 to	+0.423	+0.122	30

—●— Median randomized HD-sEMG-like estimate (10 MUs) -△- Mean of all eligible MUs
 Central 95% across-selection-seed range - - - Simulation truth (all active MUs)

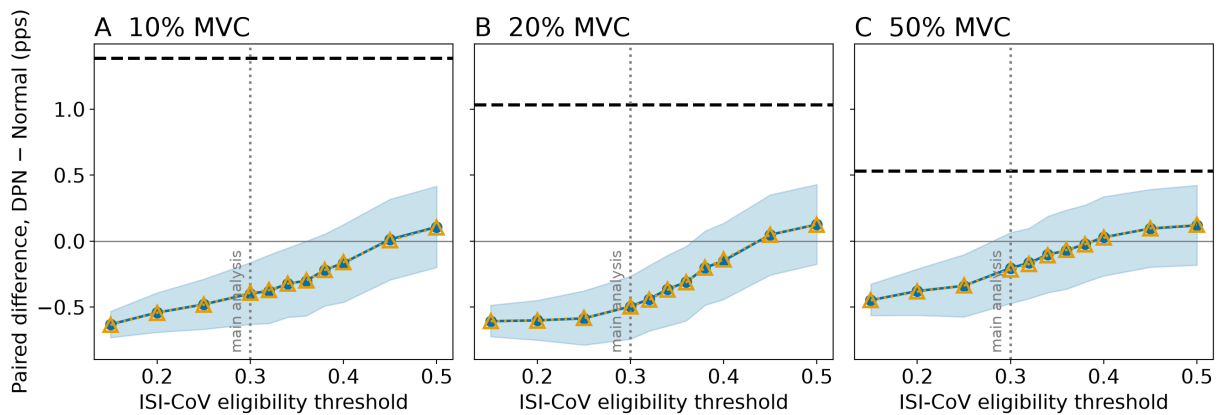


Figure saved to: diabetes/figures/selection_threshold_sensitivity_v2.png

Sweep saved to: diabetes/csv_results/selection_threshold_sensitivity_v2.csv